



MERN Full Stack Final Assessment (FA)

Set A — Placement Recruitment System

API Response & Frontend Evaluator Contract

1. Purpose of This Document

This document defines the:

- expected API response formats
- status codes
- pagination structures
- authentication response structures
- analytics response formats
- frontend evaluator conventions
- Playwright compatibility requirements

for the:

Placement Recruitment System — Set A

Students must strictly follow:

- route names
- response structures
- status codes
- data-testid conventions
- window.appState structure

mentioned in this document.

Failure to follow these conventions may affect automated evaluation.

2. Common Backend Response Standards

Success Response

Status Code

200 / 201

Expected Structure

```
{
  "success": true,
  "message": "Operation successful",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```

Error Response

Status Codes

Scenario	Status
Validation Error	400
Unauthorized	401
Forbidden	403
Not Found	404
Server Error	500

Expected Structure

```
{
  "success": false,
  "message": "Error message"
}
```

Pagination Response Format

Expected Structure

```
{
  "success": true,
  "message": "Data fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

3. MongoDB & Authentication Routes

POST /sync

Status Code

200

Auth Required

Yes

Allowed Roles

admin

placement_officer

Expected Response

```
{
  "success": true,
  "message": "Dataset synchronized successfully",
  "data": {
    "totalFetched": 100,
    "inserted": 82,
    "duplicates": 5,
    "rejected": 13
  }
}
```

Values are dynamic based on dataset.

GET /health

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Database connected successfully",
  "data": {
    "database": "connected",
    "documentCount": 82
  }
}
```

POST /auth/register

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "User registered successfully",
  "data": {
    "_id": "string",
    "name": "string",
    "email": "string",
    "role": "string"
  }
}
```

```
}  
}
```

POST /auth/login**Status Code**

200

Expected Response

```
{  
  "success": true,  
  "message": "Login successful",  
  "token": "JWT_TOKEN",  
  "data": {  
    "_id": "string",  
    "name": "string",  
    "email": "string",  
    "role": "string"  
  }  
}
```

GET /auth/me**Status Code**

200

Auth Required

Yes

Expected Response

```
{  
  "success": true,  
  "message": "Authenticated user fetched successfully",  
  "data": {  
    "_id": "string",  
    "name": "string",  
    "email": "string",  
    "role": "string"  
  }  
}
```

4. Student APIs**GET /students****Status Code**

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Students fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string",
      "studentId": "string",
      "name": "string",
      "email": "string",
      "department": "string",
      "cgpa": 8.5,
      "status": "active"
    }
  ]
}
```

GET /students/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Student fetched successfully",
  "data": {
    "_id": "string",
    "studentId": "string",
    "name": "string",
    "email": "string",
    "department": "string",
    "cgpa": 8.5,
    "skills": [],
    "graduationYear": 2027,
    "phone": "string",
  }
}
```

```
"status": "active"
}
```

5. Company APIs

POST /companies

Status Code

201

Auth Required

Yes

Allowed Roles

admin

placement_officer

Expected Response

```
{
  "success": true,
  "message": "Company created successfully",
  "data": {
    "_id": "string",
    "companyId": "string",
    "name": "string",
    "role": "string"
  }
}
```

GET /companies

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Companies fetched successfully",
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

**GET /companies/:id****Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Company fetched successfully",
  "data": {
    "_id": "string",
    "companyId": "string",
    "name": "string",
    "role": "string",
    "package": 8
  }
}
```

PATCH /companies/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Company updated successfully",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /companies/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Company deleted successfully"
}
```

6. Drive APIs

POST /drives

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Drive created successfully",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```

GET /drives

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Drives fetched successfully",
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

GET /drives/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Drive fetched successfully",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```



```
"updatedAt": "ISO_DATE"
}
```

PATCH /drives/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Drive updated successfully",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /drives/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Drive deleted successfully"
}
```

7. Application APIs

POST /applications

Status Code

201

Expected Response

```
{
  "success": true,
  "message": "Application submitted successfully",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
```

```
"updatedAt": "ISO_DATE"
}
```

GET /applications

Status Code

200

Pagination Supported

Yes

Expected Response

```
{
  "success": true,
  "message": "Applications fetched successfully",
  "page": 1,
  "limit": 10,
  "total": 50,
  "totalPages": 5,
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

GET /applications/:id

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Application fetched successfully",
  "data": {
    "_id": "string",
    "applicationId": "string",
    "student": {},
    "drive": {},
    "status": "shortlisted"
  }
}
```

**PATCH /applications/:id****Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Application updated successfully",
  "data": {
    "_id": "string",
    "createdAt": "ISO_DATE",
    "updatedAt": "ISO_DATE"
  }
}
```

DELETE /applications/:id**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Application deleted successfully"
}
```

8. Interview APIs**PATCH /
interviews/:id Status****Code**

200

Allowed Results

pending

pass

fail

Expected Response

```
{
  "success": true,
  "message": "Interview result updated successfully",
  "data": {
    "result": "pass"
  }
}
```

```
}
```

Workflow Validation Rules

- rejected applications cannot receive interviews
- selected candidates cannot be rescheduled

9. Filtering, Search & Pagination APIs

GET /students?department=CSE

GET /students?cgpaMin=8

GET /students?status=active

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Students filtered successfully",
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

GET /drives?status=open

GET /drives?company=TechNova

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Drives filtered successfully",
  "data": [
    {
      "_id": "string"
    }
  ]
}
```



GET /applications?page=1&limit=10

GET /applications?search=TechNova

GET /applications?status=shortlisted&page=2&limit=5

Status Code

200

Expected Pagination Response

```
{
  "success": true,
  "message": "Applications fetched successfully",
  "page": 2,
  "limit": 5,
  "total": 50,
  "totalPages": 10,
  "data": [
    {
      "_id": "string"
    }
  ]
}
```

10. Analytics APIs

GET /analytics/

placements Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Placement analytics fetched successfully",
  "data": {
    "totalApplications": 100,
    "shortlisted": 40,
    "selected": 20,
    "rejected": 15
  }
}
```

Values are dynamic.

GET /analytics/departments

Status Code

200

Expected Response

```
{
  "success": true,
  "message": "Department analytics fetched successfully",
  "data": [
    {
      "department": "CSE",
      "placedStudents": 25,
      "placementPercentage": 82
    }
  ]
}
```

Values are dynamic.

GET /analytics/companies**Status Code**

200

Expected Response

```
{
  "success": true,
  "message": "Company analytics fetched successfully",
  "data": [
    {
      "company": "TechNova",
      "selectedStudents": 12,
      "highestPackage": 12,
      "driveParticipationCount": 3
    }
  ]
}
```

Values are dynamic.

11. Frontend Evaluator Contract

Mandatory React Router Routes

Route	Purpose
/login	Authentication
/dashboard	Analytics dashboard
/students	Student management

/companies	Company management
/drives	Placement drives
/applications	Application management
/interviews	Interview workflows
/profile	Current user profile

Mandatory window.appState Structure

Students must expose:

window.appState

Expected Structure

```
window.appState =  
  { authUser,  
    token,  
    students,  
    companies,  
    drives,  
    applications,  
    interviews,  
    filters,  
    analytics  
  }
```

Runtime State Validation

The evaluator may validate state updates dynamically after:

- POST operations
- PUT operations
- PATCH operations
- DELETE operations
- login/logout actions
- filtering/search actions
- pagination actions